Lab Mid Exam: HOG vs SIFT on Blood-Cell Images

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My settings: seed 56, HOG cell 8 x 8, SIFT grid 6 x 6 (36 keypoints)

Q1. Data Setup

Train: 400 images (100 per class). Validation: 120 images (30 per class). Test: 120 images (30 per class). Classes: eosinophil, monocyte, neutrophil, platelet. All images are grayscale, 128 x 128.

Q2. HOG + SVM

Feature length (by hand)

Cells per side = 128 / 8 = 16.
Blocks per side = 16 - 1 = 15, so 15 x 15 = 225 blocks.
Each block = 2 x 2 x 9 = 36 values.
Feature length = 225 x 36 = **8100**.

Result (validation)

C values tried: 0.1, 1, 10. All gave validation accuracy 0.725, so I kept C = 0.1.

Validation accuracy = 0.7250. Macro-F1 = 0.7166.

```
Rows = true, columns = predicted
(eosinophil, monocyte, neutrophil, platelet)
[[13  7  8  2]
 [ 5 22  3  0]
 [ 5  3 22  0]
 [ 0  0  0 30]]
```

Q3. SIFT + SVM

Normal SIFT keypoints (train images)

Average = 72.28, minimum = 10, maximum = 177. Images with zero keypoints = 0 out of 400.

Different keypoint counts are a problem because an SVM needs the same feature length for every image, and normal SIFT gives a different number of descriptors for each image.

Fixed grid (by hand)

6 x 6 = 36 grid points. Each point gives 128 values.
Feature length = 36 x 128 = **4608**.

Result (validation)

C values tried: 0.1, 1, 10 gave validation accuracy 0.8417, 0.8500 and 0.8083, so I chose C = 1.

Validation accuracy = 0.8500. Macro-F1 = 0.8400.

```
Rows = true, columns = predicted
(eosinophil, monocyte, neutrophil, platelet)
[[16  8  4  2]
 [ 2 28  0  0]
 [ 1  1 28  0]
 [ 0  0  0 30]]
```

Q4. Robustness

Prediction (written before the test)

Rotation: I think HOG will lose more, because its orientation bins and cell positions change when the image turns. Noise: I think HOG will lose more, because noise adds many small random gradients, while SIFT smooths the patch first.

Test results

Model	Clean accuracy	Rotated accuracy	Noisy accuracy
HOG + SVM	0.7000	0.6917	0.4000
SIFT + SVM	0.7917	0.8500	0.7917

Was my prediction correct?

Yes for noise: HOG dropped from 0.7000 to 0.4000, while SIFT stayed at 0.7917. For rotation, HOG dropped a little (0.7000 to 0.6917) and SIFT went up (0.7917 to 0.8500), so HOG lost more. But the rotation change is small for both models (about 1 and 7 images out of 120), so I cannot say rotation really hurt either one.

Q5. Short Answers and AI Log

(a) Why does HOG use block normalisation?

Gradient strength changes with lighting and contrast. Normalising each block makes the features depend on shape and edge direction, not on brightness or contrast.

(b) Why is it wrong to choose C using the test set?

Then the test set is no longer unseen, and the accuracy looks better than it really is. C should be chosen on the validation set, and the test set used only once at the end.

Most confused classes

HOG confused eosinophil and neutrophil most (13 mistakes). SIFT confused eosinophil and monocyte most (10 mistakes). An eosinophil has red-pink cytoplasm, but in grayscale this colour becomes a normal gray, so only the nucleus shape and size are left, and these look similar in all three cell types.

AI log

Prompt	What I changed	How I checked
Make a CPU-only notebook for HOG + SVM vs SIFT + SVM on BloodMNIST using my registration number.	I added my own conditions: dataset from a direct link, kept in RAM, CPU only. I saved the figures in my assets folder myself.	Ran it in Colab, checked the printed settings against my registration number, compared feature lengths (8100 and 4608) with my hand calculation, and checked each confusion matrix row adds to 30.
The download cell gives HTTP Error 403, fix it.	Replaced the download cell with one that tries other headers first and then medmnist.	Ran it again in Colab. The dataset loaded and the shapes and class counts printed correctly.